Automated Student Performance Prediction Using AutoML

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***Abstract*—The early identification of students at academic risk is an important task in educational data mining, as it enables timely academic support and informed decision-making. This paper presents an automated student performance prediction framework using machine learning techniques. The proposed approach utilizes the Student Performance Mathematics dataset from the UCI Machine Learning Repository and formulates student performance prediction as a binary classification problem, categorizing students into At Risk and Performing Well groups. A preprocessing pipeline is developed to handle numerical and categorical attributes using standardization and one-hot encoding. A Gradient Boosting classifier is employed as the baseline model, followed by hyperparameter optimization using RandomizedSearchCV with stratified five-fold cross-validation and balanced accuracy as the optimization metric. The baseline model achieves an accuracy of 89.87**

***Index Terms*—Educational Data Mining, Student Performance Prediction, Machine Learning, Gradient Boosting, Early Risk Detection, Academic Analytics**

I. Introduction

Student performance prediction is an important research area in Educational Data Mining (EDM), as it helps educational institutions identify students who may experience academic difficulties. Early identification of at-risk students can support timely academic intervention and improve educational decision-making. Machine learning techniques provide an effective approach for analyzing student attributes and predicting academic outcomes.

This paper presents an automated student performance prediction framework using the Student Performance Mathematics dataset from the UCI Machine Learning Repository. The task is formulated as a binary classification problem, categorizing students as At Risk or Performing Well based on their academic and other attributes. A preprocessing pipeline is developed to handle numerical and categorical features, and a Gradient Boosting classifier is employed for prediction. RandomizedSearchCV with stratified five-fold cross-validation is used for hyperparameter optimization.

The study evaluates the baseline and tuned models using accuracy, balanced accuracy, precision, recall, F1-score, and confusion matrix analysis. Particular emphasis is placed on recall for the At Risk class to assess the model's ability to identify students who may require academic support.

The main contributions of this work are:

- Development of a machine learning framework for binary student performance classification.
- Implementation of a preprocessing pipeline for numerical and categorical attributes.
- Comparative evaluation of baseline and hyperparameter-tuned Gradient Boosting models.
- Analysis of at-risk student identification using class-specific evaluation metrics.

The remainder of this paper is organized as follows. Section II presents related work. Section III describes the proposed methodology. Section IV discusses the experimental setup, Section V presents the results, and Section VI concludes the paper.

II. Related Work

Educational Data Mining (EDM) applies data mining and machine learning techniques to analyze student learning patterns and academic outcomes. Student performance prediction is an important area of EDM, as it helps identify students who may face academic difficulties and enables timely academic support. Various machine learning algorithms, including Decision Trees, Random Forests, Support Vector Machines, and ensemble methods, have been explored for predicting academic achievement.

Previous studies have utilized academic records, demographic information, and behavioral attributes to predict student performance. These attributes provide valuable insights into students' academic progress and learning patterns. However, the effectiveness of prediction models depends on the dataset, preprocessing techniques, and selected algorithms. In addition, overall accuracy may not sufficiently indicate a model's ability to identify at-risk students.

Therefore, this study focuses on binary classification of student performance using the UCI Student Performance Mathematics dataset. A Gradient Boosting classifier is implemented with a preprocessing pipeline for numerical and categorical features. The baseline and hyperparameter-tuned models are evaluated using multiple performance metrics, with particular attention to identifying students who may require academic support. Table I summarizes the existing studies, their limitations, and their relevance to the present work.

TABLE I
LITERATURE REVIEW OF EXISTING STUDIES

S.No.	Author, Conference, Year	Algorithm Used	Key Methodology	Limitation	Research Gap	How It Helps Our Project
1	P. Umamaheswari et al., ICMAT23, 2024	Modified SVM, KNN	Classifies and predicts student success using student data. The modified SVM is evaluated using accuracy, recall, and F1-score.	Depends on the available dataset and selected features.	No automated algorithm selection or hyperparameter tuning.	Provides SVM and KNN as candidate models for our AutoML pipeline.
2	S. Paddalwar et al., ICACC-2022, 2022	ML with Hybrid Feature Selection	Combines multiple feature-selection techniques with machine learning algorithms for academic-grade prediction.	Depends on the quality and relevance of selected features.	Fully automated feature selection and model optimization can be explored.	Supports automated feature selection and model selection in our system.
3	T. S. Franciskovic et al., CSEDU, 2024	ML, Regression, Clustering	Uses pre-class engagement and learning-trace data to predict final examination scores.	Based on a specific flipped-classroom environment.	More diverse attributes and automated model optimization are needed.	Shows that engagement and learning activity can be useful prediction features.
4	H. Alrasheed et al., DSDE, 2021	Statistical and Correlation Analysis	Studies the relationship between English proficiency and academic performance among software engineering students.	Focuses mainly on English proficiency and a specific student group.	Multiple academic, behavioral, and demographic features can be combined.	Supports the inclusion of diverse student attributes in prediction.
5	M. S. Sallam et al., IICC, 2026	ML Models, XAI	Predicts student adaptability using machine learning models and XAI techniques to understand prediction factors.	Focuses on adaptability rather than direct academic-grade prediction.	Broader student-performance indicators and automated optimization can be explored.	Supports model comparison and explainable prediction in our system.
6	V. Balasubramanian et al., CSEDU, 2024	SVM, XGBoost	Applies preprocessing and feature engineering followed by SVM and XGBoost for academic-outcome prediction.	Uses a limited dataset and manually selected algorithms.	Lacks end-to-end automation of model selection and hyperparameter optimization.	Provides SVM and XGBoost as candidate models for our AutoML pipeline.
7	B. R. Pandian et al., 2024	Classification, Regression, Ensemble Learning	Reviews machine learning techniques, datasets, predictive factors, model performance, and interpretability for student prediction.	Primarily a literature review rather than an implemented framework.	Need for reliable, explainable, and automated ML systems.	Supports the motivation for automated model selection and optimization.
8	Z. Xu, CSPML, 2024	AI/ML, Data Analytics	Investigates AI and ML applications in education, learning analytics, and personalized learning.	Broad AI-in-education scope; prediction is not the sole focus.	Need for focused and automated student-performance prediction.	Provides broader motivation for automated educational data analytics.
9	B. K. Verma et al., ICCSAI, 2025	Clustering, Data Mining	Uses clustering-guided data mining to identify patterns in hybrid-learning environments.	Focuses mainly on clustering rather than supervised prediction.	Clustering can be combined with automated supervised prediction.	Can support exploratory analysis and student grouping before AutoML prediction.
10	S. Agustri et al., ICIMCIS, 2023	ML Recommender Systems	Reviews recommender systems using student information and performance-related data.	Focuses on recommendation rather than direct performance prediction.	Prediction can be connected with personalized recommendations.	Our system can potentially provide recommendations for identified at-risk students.

A. Research Gap and Motivation

Existing studies have explored various machine learning techniques for predicting student performance. However, differences in datasets, preprocessing methods, and model configurations make their results difficult to compare. The need for systematic model evaluation and hyperparameter optimization remains an important research area.

Furthermore, overall accuracy may not adequately reflect a model's ability to identify at-risk students. Therefore, class-specific metrics such as precision, recall, and F1-score are important for evaluating prediction performance. Identifying students who may experience academic difficulties can help educational institutions provide timely and appropriate academic support.

To address these gaps, this study develops an Automated Student Performance Prediction framework using the UCI Student Performance Mathematics dataset. The framework classifies students into At Risk and Performing Well categories based on their academic and other attributes. A Gradient Boosting classifier with a preprocessing pipeline is evaluated before and after hyperparameter optimization. The study also examines class-specific performance to assess the model's effectiveness in identifying students who may require academic support. The evaluation also provides insights into the strengths and limitations of the selected model. The results can help support data-driven academic decisions and guide timely interventions for students who need additional assistance.

III. Proposed Methodology

The proposed framework uses machine learning techniques to predict student academic performance and identify students who may require academic support. The methodology consists of data collection, preprocessing, model development, hyperparameter optimization, and performance evaluation. The UCI Student Performance Mathematics dataset is used to develop a binary classification model that categorizes students as At Risk or Performing Well. The overall workflow of the proposed framework is illustrated in Fig. 1

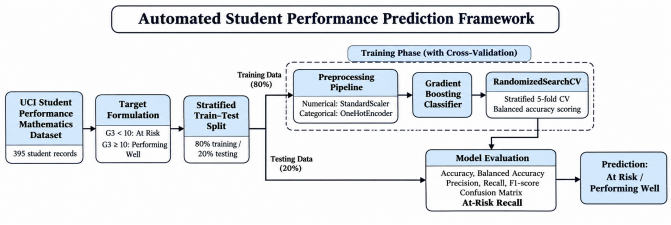


Fig. 1. Architecture of the proposed Automated Student Performance Prediction framework.

A. Dataset Description

The proposed framework uses the Student Performance Mathematics dataset obtained from the UCI Machine Learning Repository. The dataset contains 395 student records with 33 attributes, including demographic information, family background, and academic characteristics. These attributes provide information about students' academic performance and related factors.

The final mathematics grade, represented by G3, is used to define the target variable. Students with G3 below 10 are categorized as At Risk, while students with G3 greater than or equal to 10 are categorized as Performing Well. This transforms the original grade prediction task into a binary classification problem.

The dataset is divided into training and testing subsets using an 80:20 stratified split. Stratification preserves the class distribution in both subsets, allowing the model to be trained and evaluated consistently.

B. Data Preprocessing

Data preprocessing is performed to prepare the dataset for machine learning. The input features are separated into numerical and categorical attributes. Numerical features are standardized using StandardScaler, while categorical features are transformed using OneHotEncoder with unknown categories ignored during transformation.

A ColumnTransformer is used to apply these preprocessing operations to the corresponding feature groups. The preprocessing pipeline is integrated with the Gradient Boosting classifier to ensure that the transformations are consistently applied during training and testing. The target variable is excluded from the input features to avoid data leakage.

C. Model Development

Gradient Boosting is an ensemble machine learning algorithm that combines multiple weak learners, typically decision trees, to form a strong predictive model. The trees are built sequentially, where each new tree attempts to correct the errors made by the previous trees. This iterative learning process helps the model capture complex relationships between input features and the target variable.

In this study, the Gradient Boosting classifier is trained using the preprocessed student attributes to predict academic performance. The model classifies students into At Risk and Performing Well categories based on their academic and other characteristics. Its performance is evaluated using an independent testing dataset to assess its ability to predict student outcomes.

D. Hyperparameter Optimization

Hyperparameter optimization is performed to identify a suitable configuration for the Gradient Boosting classifier. RandomizedSearchCV is used to explore different combinations of hyperparameter values. The search is conducted using stratified five-fold cross-validation, which maintains the class distribution across the folds.

Balanced accuracy is used as the scoring metric to account for the performance of both classes. The hyperparameters considered include the learning rate, maximum tree depth, number of estimators, minimum samples per leaf, and subsampling ratio. The configuration achieving the highest mean cross-validation balanced accuracy is selected for evaluation on the independent test set.

E. Performance Evaluation

The performance of the proposed student performance prediction model is evaluated using multiple classification metrics. These metrics provide a comprehensive understanding of the model's predictive ability and its effectiveness in identifying students who may require academic support. Since the dataset contains two classes, At Risk and Performing Well, evaluating the performance of each class is important.

The evaluation metrics are calculated using the terms True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). These terms represent correctly and incorrectly classified instances with respect to the positive class. In this study, the At Risk category is considered the positive class for class-specific evaluation.

Accuracy: Accuracy measures the proportion of correctly classified instances among all predictions. It provides an overall measure of the model's predictive performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision: Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive. It indicates the reliability of the model's positive predictions.

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

Recall: Recall measures the proportion of actual positive instances correctly identified by the model. In this study, recall for the At Risk class is particularly important because it indicates how effectively the model identifies students who may need academic support.

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

F1-Score: The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of these two metrics.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

Balanced Accuracy: Balanced accuracy is calculated as the average recall obtained for each class. It provides a class-balanced measure of predictive performance.

$$BalancedAccuracy = \frac{Recall_{AtRisk} + Recall_{PerformingWell}}{2} \tag{5}$$

A confusion matrix is also used to examine the number of correctly and incorrectly classified instances for each class. The evaluation metrics are calculated on the independent test dataset, which is not used during model training or hyperparameter optimization. The baseline and tuned models are compared using the same test dataset to understand the effect of hyperparameter optimization on predictive performance.

IV. Results and Discussion

This section presents the experimental results obtained from the proposed student performance prediction framework. The performance of the baseline Gradient Boosting classifier is analyzed using the held-out test dataset. The evaluation considers overall classification performance, class-wise prediction metrics, and the ability of the model to identify students who may require academic support. The results are further analyzed to understand the effectiveness of the proposed approach.

A. Experimental Setup

The experiments are conducted using the UCI Student Performance Mathematics dataset, which contains 395 student records. The target variable is generated from the final mathematics grade (G3), where students with grades below 10 are categorized as At Risk and students with grades greater than or equal to 10 are categorized as Performing Well.

The dataset is divided into training and testing subsets using an 80:20 stratified split with a random state of 42. The training subset contains 316 records, while the testing subset contains 79 records. The Gradient Boosting classifier is trained using the preprocessed input features. RandomizedSearchCV with stratified five-fold cross-validation is used for hyperparameter optimization, with balanced accuracy as the scoring metric.

The baseline and hyperparameter-tuned models are evaluated on the same independent test dataset. This experimental setup enables a consistent comparison of their predictive performance.

B. Baseline Model Performance

The baseline Gradient Boosting classifier was evaluated using the held-out test dataset containing 79 student records. The model achieved an accuracy of 89.87% and a balanced accuracy of 91.47%. The macro F1-score was 0.891, while the weighted F1-score was 0.901. These results indicate that the model provides effective classification performance across the two student categories.

For the At Risk class, the model achieved a precision of 78.12%, a recall of 96.15%, and an F1-score of 0.862. The high recall indicates that the classifier successfully identified 25 out of 26 students who were actually at risk. However, one at-risk student was incorrectly classified as Performing Well, which could result in a student requiring academic support being overlooked.

For the Performing Well class, the model achieved a precision of 97.9%, a recall of 86.8%, and an F1-score of 0.920. Out of 53 students belonging to this category, 46 were correctly classified, while seven were predicted as At Risk. Although these predictions may lead to some students being incorrectly flagged for support, they may also encourage further academic assessment.

Overall, the baseline model demonstrates strong performance in identifying at-risk students. However, the difference between precision and recall indicates that the model tends to identify more students as At Risk than the actual number in this category. The confusion matrix provides a detailed representation of these classification outcomes.

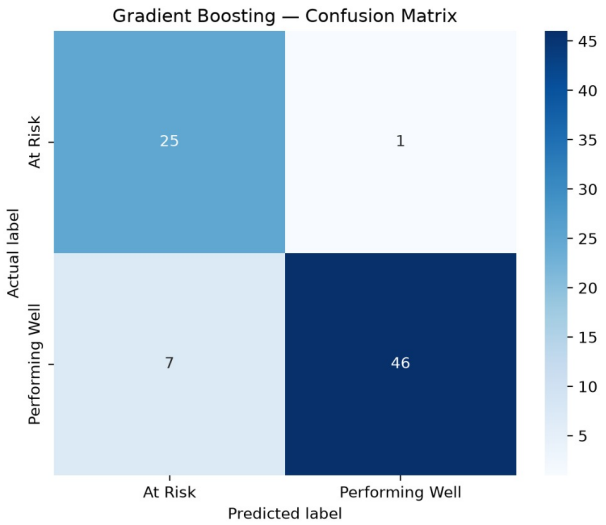


Fig. 2. Confusion matrix of the baseline Gradient Boosting classifier on the held-out test dataset.

TABLE II
CLASSIFICATION REPORT OF THE BASELINE GRADIENT BOOSTING MODEL

Class	Precision	Recall	F1-Score
At Risk	0.781	0.962	0.862
Performing Well	0.979	0.868	0.920
Macro Average	0.880	0.915	0.891
Weighted Average	0.914	0.899	0.901

C. Hyperparameter-Tuned Model Performance

Hyperparameter optimization was performed using RandomizedSearchCV with stratified five-fold cross-validation. Balanced accuracy was used as the scoring metric to identify a suitable configuration for the Gradient Boosting classifier. The best configuration achieved a mean cross-validation balanced accuracy of 0.9209.

The selected hyperparameters included a learning rate of 0.1305, a maximum tree depth of 2, 147 estimators, a minimum of 2 samples per leaf, and a subsampling ratio of 0.8112. The tuned model was subsequently evaluated on the same independent test dataset used for the baseline model.

The tuned classifier achieved an accuracy of 88.61%, a balanced accuracy of 89.55%, and a macro F1-score of 0.8765. The corresponding confusion matrix was:

$$\begin{bmatrix} 24 & 2 \\ 7 & 46 \end{bmatrix} \tag{6}$$

Compared with the baseline model, the tuned classifier showed a slight reduction in test accuracy and balanced accuracy. The baseline model achieved an accuracy of 89.87% and a balanced accuracy of 91.47%, whereas the tuned model achieved 88.61% and 89.55%, respectively. This indicates that the hyperparameter optimization process improved the cross-validation score but did not improve the performance on the independent test dataset.

D. Feature Importance Analysis

Feature importance analysis is performed to understand the contribution of input attributes to the predictions made by the Gradient Boosting classifier. Figure 3 presents the top 15 features based on their importance scores.

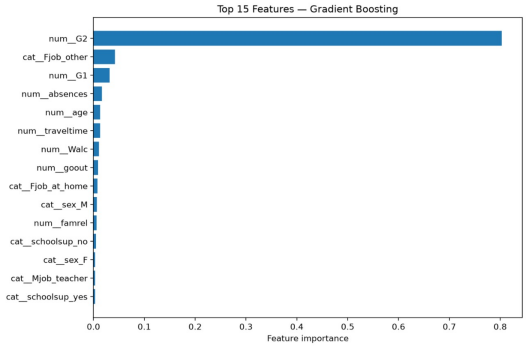


Fig. 3. Top 15 feature importance scores of the Gradient Boosting classifier.

The results indicate that G2 is the most influential feature, with an importance score of 0.8031. This suggests that the student's second-period grade plays a major role in predicting final academic performance.

E. Discussion

The experimental results demonstrate that the Gradient Boosting classifier can effectively predict student performance using academic and demographic attributes. The baseline model achieved an accuracy of 89.87% and a balanced accuracy of 91.47% on the independent test dataset.

The confusion matrix shows that 25 out of 26 actual At Risk students were correctly identified, resulting in an At Risk recall of 96.15%. However, one At Risk student was incorrectly classified as Performing Well. This highlights the importance of evaluating class-specific recall, as incorrectly identifying an at-risk student may delay academic support.

Feature importance analysis indicates that G2, the student's second-period grade, is the most influential feature, with an importance score of 0.8031. G1, absences, and other student attributes also contribute to the predictions. The strong influence of G2 suggests that previous academic performance is an important indicator of final performance.

Although hyperparameter optimization achieved a cross-validation balanced accuracy of 92.09%, the tuned model did not improve the independent test results. The baseline model performed slightly better on the test dataset, indicating that a higher cross-validation score does not necessarily translate into better generalization. Overall, the results show the potential of the proposed framework as a decision-support tool for identifying students who may require additional academic assistance.

V. CONCLUSION

This study presented an Automated Student Performance Prediction framework using the UCI Student Performance Mathematics dataset. A Gradient Boosting classifier was developed to categorize students into At Risk and Performing Well groups. The framework incorporated data preprocessing, stratified data splitting, and hyperparameter optimization using RandomizedSearchCV.

The baseline model achieved an accuracy of 89.87% and a balanced accuracy of 91.47%, with an At Risk recall of 96.15%. Although hyperparameter optimization achieved a cross-validation balanced accuracy of 92.09%, the tuned model did not improve performance on the independent test dataset. Feature importance analysis identified G2 as the most influential feature, followed by other academic and student-related attributes.

The results demonstrate the potential of machine learning for identifying students who may require academic support. Future work will explore additional machine learning algorithms, larger and more diverse datasets, and improved model optimization techniques to enhance prediction reliability and support timely educational interventions.

REFERENCES

[1] P. Umamaheswari, M. Vanitha, P. Vimala Devi, J. Glory Thephoral, and B. Rihan Basha, "Student success prediction using a novel machine learning approach based on modified SVM," *Multidisciplinary Science Journal*, vol. 6, 2024, Art. no. e2024ss0110, doi: 10.31893/multiscience.2024ss0103.

[2] S. Paddalwar, V. Mane, and L. Ragha, "Predicting students' academic grade using machine learning algorithms with hybrid feature selection approach," *ITM Web of Conferences*, vol. 44, Art. no. 03036, 2022, doi: 10.1051/itmconf/20224403036.

[3] T. S. Franciskovic, A. Andelic, J. Slivka, N. Luburic, and A. Kovacevic, "Predicting students' final exam scores based on their regularity of engagement with pre-class activities in a flipped classroom," in *Proc. 16th Int. Conf. on Computer Supported Education (CSEDU)*, 2024, pp. 97–107, doi: 10.5220/0012681800003693.

[4] H. Alrasheed, A. Alnashwan, and R. Alshowiman, "Impact of English proficiency on academic performance of software engineering students," in *Proc. 4th Int. Conf. on Data Storage and Data Engineering (DSDE)*, 2021, pp. 107–111, doi: 10.1145/3456146.3456163.

[5] M. S. Sallam, K. Metwally, and K. S. Badran, "Predicting student adaptability level using ML models and XAI approaches," in *Proc. ICEENG International Conference for Innovations in Intelligent Computing and Cybersecurity (IICC)*, 2026.

[6] V. Balasubramanian et al., "[Insert the exact paper title]," in *Proc. 16th Int. Conf. on Computer Supported Education (CSEDU)*, 2024.

[7] B. R. Pandian, A. A. Aziz, H. Subramaniam, and H. S. A. Nawi, "Exploring the role of machine learning in forecasting student performance in education: An in-depth review of literature," *Multidisciplinary Reviews*, vol. 6, Art. no. 2023ss043, 2024, doi: 10.31893/multirev.2023ss043.

[8] Z. Xu, "AI in education: Enhancing learning experiences and student outcomes," in *Proc. 4th Int. Conf. on Signal Processing and Machine Learning (CONF-SPML)*, 2024, doi: 10.54254/2755-2721/51/20241187.

[9] B. K. Verma, N. Srivastava, and A. K. Bharti, "Optimizing hybrid learning outcomes via clustering-guided data mining techniques in higher education," in *Proc. 3rd Int. Conf. on Communication, Security, and Artificial Intelligence (ICCSAI)*, 2025, doi: 10.1109/ICCSAI64074.2025.11064577.

[10] S. Agustri, E. Ermatita, and D. P. Rini, "Recommender systems for students' performance in higher educations: A review," in *Proc. 2023 Int. Conf. on Informatics, Multimedia, Cyber and Information Systems (ICIMCIS)*, 2023, doi: 10.1109/ICIMCIS60089.2023.10348622.